Department of Computer Science & Engineering

PERSONAL BUDGET & EXPENSE SPLITTER (SMARTSPLIT)

MINI PROJECT | SESSION 2026-2027

PRESENTED BY : GROUP NO. 7

1. **Pratibha Singh** (Roll: 2501920100419)
2. **Pratiksha Singh** (Roll: 2501920100422)
3. **Priyansh Kalra** (Roll: 2501920100438)
4. **Priyam Tripathi** (Roll: 2501920100437)

2nd Year B.Tech CSE | 4th Semester

UNDER SUPERVISION OF :

Project Guide:
Mr. Abhijeet Gupta
Designation: Assistant Professor
Department: Computer Science & Engineering
G. L. Bajaj Institute of Technology & Management, Greater Noida
Affiliated with Dr. A.P.J. Abdul Kalam Technical University (AKTU)

GL BAJAJ Institute of Technology & Management
FIND YOUR SPARK

OUTLINE

01 Introduction

02 Objectives & Scope

03 Existing vs Proposed System

04 Technology Stack

05 Resource Requirements

06 System Design & Architecture

07 Methodology

08 Modules & Features

09 Greedy Settlement Algorithm

10 Implementation & Results

11 Testing & Validation

12 Conclusion & References

INTRODUCTION

01 Domain & Background

- FinTech & Distributed Web Systems engineering domain.
- Shared living expenses are routine for college roommates, project groups, and traveling peers.
- Manual calculation creates confusion, misremembered debts, and social friction.

02 Problem Statement

- Messy paper logs and group chat messages lead to missed records and forgotten settlements.
- Circular debts result in unnecessary multiple peer transactions (A owes B, B owes C, C owes A).
- Existing market apps lock vital settlement features behind paid monthly subscriptions.

03 Motivation & Significance

- Build a fast, ad-free, transparent platform specifically tailored for students and shared groups.
- Algorithmic debt optimization reducing multi-party payments to minimal direct transfers.
- Native UPI payment links and instant real-time synchronization via Google Firestore.

OBJECTIVES & SCOPE

Project Objectives

- **Design & Develop** a clean, responsive single-page web app for tracking individual & group budgets.
- **Implement Multi-Type Splits** supporting Equal, Exact Amounts, Percentage, and Custom Share weights.
- **Formulate Greedy Debt Simplification** running in O(N log N) to eliminate cyclic liabilities.
- **Integrate Dynamic UPI QR Codes** for direct seamless peer settlements without manual VPA entry.
- **Ensure Secure Persistence** with Firebase Firestore database security rules and token authentication.

Project Scope Boundaries

DELIVERED IN VERSION 1.0 (INCLUDED):

• Full Group Lifecycle, Multi-currency (INR ₹, USD $, EUR €, GBP £), In-App Group Chat, Monthly Analytics, and Demo Sandbox Mode.

FUTURE SCOPE & LIMITATIONS (VERSION 2.0):

• Automated OCR Bill scanning via Vision API, direct banking webhook reconciliation, and packaging as a native Android APK.

EXISTING VS PROPOSED SYSTEM

Evaluation Parameter	Manual / Excel Sheets	Commercial Split Apps	SmartSplit (Proposed)
Debt Simplification	None (Raw totals)	Basic (Locked in Pro)	**O(N log N) Greedy (Auto & Free)**
UPI QR Payment	Manual copy-paste of VPA	External redirect	**Direct Dynamic UPI QR Generated**
Split Modalities	Equal only	Equal & Percentage	**Equal, Exact, %, Shares, Unequal**
In-App Chat	External messaging app	Comment threads only	**Dedicated Real-time Group Chat Channel**
Cost & Privacy	Manual file sharing	Ad-heavy / Subscription tier	**100% Free & Open Source Sandbox**

Literature Survey Insight: Existing tools leave a significant gap between simple calculators and cluttered commercial apps. SmartSplit solves this by combining minimal transaction optimization with native UPI payments and zero subscription barriers.

TECHNOLOGY STACK

1

Frontend Layer

React 19 & TypeScript

- React 19 Component Architecture with custom hooks.
- TypeScript for static type safety & strict schemas.
- Tailwind CSS for responsive layouts.
- Lucide React Icons for crisp interface elements.
- Vite for instant Hot-Module Replacement.

2

Backend & Logic Layer

Node.js & Algorithm Engine

- Node.js runtime environment.
- Greedy Min-Cash-Flow debt simplification.
- Deterministic UPI URI payload generation.
- Web Crypto API for secure data hashing.
- Zero-sum verification matrix validation.

3

Database & Cloud Tools

Firebase Firestore & Git

- Google Firebase Firestore NoSQL Database.
- Firebase Tokenized User Authentication.
- IndexedDB fallback for in-browser Demo Sandbox.
- Git & GitHub for version control.
- Postman for API verification.

RESOURCE REQUIREMENTS

Hardware Requirements

Processor (CPU)	Intel Core i3 / AMD Ryzen 3 or higher
RAM (Memory)	8 GB recommended (4 GB minimum)
Storage	256 GB SSD / HDD (2 GB free space)
Display	1366 x 768 or Full HD 1920 x 1080
Network	Broadband Wi-Fi connection for cloud sync

Software Requirements

Operating System	Windows 10/11, macOS, Ubuntu Linux
Code Editor / IDE	Visual Studio Code (v1.90+)
Runtime & Package	Node.js (v20 LTS) & npm
Web Browser	Chrome, Edge, Firefox (ES6+ support)
Database & Cloud	Firebase Firestore & Cloud Run

SYSTEM DESIGN & ARCHITECTURE

Tier 1: Presentation Layer (React 19 SPA & Tailwind CSS)

Interactive Dashboard, Split Wizards, Group Chat, Settlement Modals, and responsive mobile viewport.

Tier 2: Business Logic & Algorithm Layer (Node.js & Greedy Solver)

Greedy Min-Cash-Flow algorithm, multi-party split engine (equal, exact, percentage, shares), and UPI URI generator.

Tier 3: Persistence & Security Layer (Firebase Firestore & Auth)

Cloud collections (users, groups, expenses, settlements), JWT security rules, and offline IndexedDB demo sandbox.

METHODOLOGY

1

Requirement Analysis

Gather user stories, collegiate survey, and set constraints.

2

System Design

Plan UI flows, data models, and greedy algorithm logic.

3

Development

Sprint implementation of React UI and calculation engine.

4

Testing & QA

Verify zero-sum math, edge cases, and cross-browser checks.

5

Deployment

Host on Cloud Run, seed mock demo groups, and verify URLs.

Agile Methodology: Conducted 4 two-week sprint cycles with continuous integration, automated builds, and peer code reviews following Department of CSE project guidelines.

MODULES & FEATURES

01 User Authentication

Firebase tokenized login, profile customization, and guest sandbox access.

02 Smart Dashboard

Real-time summary cards: Total Owed, Total Lending, and Monthly Spending Trends.

03 Expense Splitting

Multi-party split modalities: Equal, Exact, Percentage, and Custom Share weights.

04 Group Management

Create trip/flatmate groups, invite members, set budgets, and categorize expenses.

05 Settlement & UPI QR

Automated debt reduction, settlement recording, and dynamic UPI QR generator.

06 Analytics & Export

Interactive category charts, monthly breakdown, and 1-Click PDF/Word project report export.

GREEDY SETTLEMENT ALGORITHM

Greedy Min-Cash-Flow Formulation

1. Net Balance Calculation:

For each member *i*: `Net[i] = Sum(Paid) - Sum(Owed)`. The zero-sum property guarantees: `Σ Net[i] == 0.00`.

2. Partition & Sort:

Divide into Debtors (Net < 0) and Creditors (Net > 0). Sort descending by absolute values.

3. Greedy Pairing:

Pair the largest debtor with the largest creditor. Settle `min(|Debt|, |Credit|)`. Reduces transactions from $O(N^2)$ to at most N-1!

Before vs After Optimization

4 Roommates (Trip Expense):

Pratibha paid ₹1,200 (Food) • Pratiksha paid ₹600 (Cab) • Priyansh paid ₹400 (Groceries).

Without Algorithm: 6 separate cyclic transfers required.

With SmartSplit: Just 2 direct payments!

- **Priyam pays ₹600 to Pratibha**
- **Priyansh pays ₹300 to Pratibha**

67% reduction in total transactions!

IMPLEMENTATION & RESULTS

Dashboard & Budgets

[Live Screen Verified]

Live summary cards, spending breakdown, and real-time balance stats.

Smart Split Wizard

[Live Screen Verified]

Multi-party split modalities with instant zero-sum validation.

Settlement & UPI QR

[Live Screen Verified]

One-tap payment confirmation and standard UPI QR payload.

Live App: https://ais-dev-6uq6o6ofo6pym3m3ggio77-368234083540.asia-southeast1.run.app
GitHub: Group No. 7 - Personal Budget & Expense Splitter (SmartSplit)

TESTING & VALIDATION

Test ID	Test Scenario	Input Data	Expected Result	Status
TC-01	User Authentication	Valid email & password	JWT session issued; redirect to Dashboard	**PASS**
TC-02	Equal Split Math	Amount: ₹1,000, 4 members	Exact ₹250.00 assigned per member	**PASS**
TC-03	Zero-Sum Net Balance	15 random group expenses	Sum of all Net Balances == 0.00 strictly verified	**PASS**
TC-04	Greedy Optimization	Cyclic loop: A->B->C->A	Reduced to minimal non-cyclic transactions	**PASS**
TC-05	UPI QR Payload	Payee VPA & amount ₹450	Valid standard RFC UPI payload string	**PASS**
TC-06	Latency Benchmark	50 members graph	Execution latency < 2 ms	**PASS**

Verification Summary: Completed Unit Tests on calculation engine, Functional Tests on expense wizards, and Cross-Browser tests on Chrome, Edge, and Firefox.

CONCLUSION & FUTURE SCOPE

Conclusion & Achievements

- Successfully designed and built SmartSplit for tracking and settling collegiate group expenses.
- Eliminated manual calculation mistakes and cut transaction count by over 60% with greedy optimization.
- Provided full group lifecycle management, in-app messaging, and one-tap report generation.
- Strengthened team engineering competencies across React 19, TypeScript, algorithms, and Firestore.

Future Scope & Enhancements

- **OCR Bill Parsing:** Integrate Vision AI to automatically scan printed receipts into itemized shares.
- **Direct UPI Webhooks:** Partner with payment aggregators to confirm transactions via bank webhooks.
- **Native Mobile App:** Package via Capacitor PWA for installability and instant push notifications.

REFERENCES

Academic & Technical References (APA Style)

[1] Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022). *Introduction to Algorithms* (4th ed.). MIT Press. (Greedy Min-Cash-Flow section).

[2] Facebook Open Source. (2026). *React: The Library for Web and Native User Interfaces*. Meta Platforms, Inc. https://react.dev

[3] Google Firebase. (2026). *Cloud Firestore Documentation: Real-time NoSQL Database*. Alphabet Inc. https://firebase.google.com/docs/firestore

[4] National Payments Corporation of India (NPCI). (2025). *Unified Payments Interface (UPI) Linking Specifications (v2.1)*. NPCI Technical Specifications.

[5] Mozilla Developer Network (MDN). (2026). *Web Crypto API & Client-Side Storage Guidelines*. Mozilla Foundation. https://developer.mozilla.org

[6] Dr. A.P.J. Abdul Kalam Technical University (AKTU). (2026). *B.Tech CSE Mini Project Evaluation Criteria & Format*. Lucknow, Uttar Pradesh.

All references adhere to APA 7th edition formatting guidelines as required by AKTU.

Thank You

Any questions?

Personal Budget & Expense Splitter (SmartSplit)

Group No. 7 | 2nd Year B.Tech CSE (Session 2026–2027)

Department of Computer Science & Engineering

G. L. Bajaj Institute of Technology & Management, Greater Noida